

**Effect of climatic variables on rice yield and its forecast in the North-Central
Province of Sri Lanka.**

A Research Proposal in partial fulfilment of the

requirements for the course

STAT 32543 Research Methodology, Data Analysis & Report Writing

at Department of Statistics and Computer Science, Faculty of Science, University of Kelaniya

Sri Lanka.

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1. Prefer

The research topic "Effect of climatic variables on rice yield and its forecast in the North-Central Province of Sri Lanka" was selected due to its direct relevance to the region's agricultural sustainability and food security.

Rice cultivation is a cornerstone of livelihoods in the North-Central Province, yet it faces increasing vulnerability to climate change impacts such as erratic rainfall patterns and rising temperatures.

Understanding the effect of climatic variables on rice yield is essential for implementing adaptive agricultural practices that can mitigate these risks and ensure food security for local communities. And this research helps to identify the most associated climatic factor for the rice yield in that geographical area.

Furthermore, by developing forecasting models to predict rice yield based on climatic variables, this research aims to provide valuable insights for proactive decision-making and resource allocation in agricultural management.

By anticipating fluctuations in rice production, stakeholders can better prepare for and respond to changing environmental conditions, ultimately contributing to the long-term resilience and sustainability of agriculture in the North-Central Province of Sri Lanka.

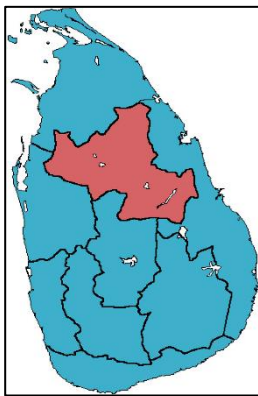


Figure 1
Representation of the North-Central Province.
By Kavindra Paranage.
Image Source; ResearchGate



Figure 2
Paddy Cultivation
Image Source; <https://nexttravelsrilanka.com>

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5. Introduction

a) Statement of the Problem in Laymen Term

- For a large proportion of the world's population, rice is a staple diet, and weather patterns have a big impact on how much is produced. It is essential to comprehend how climate factors affect the yield of rice to maintain sustainable agriculture and food security.
- So, the goal of this study is to predict rice yield in North-Central Province of Sri-Lanka, based on weather factors like temperature, rainfall, humidity, and sunshine hours by analyzing annual historical data. And, to identify the most associated climatic factor for the rice yield.
- Furthermore, the study aims to offer insights that can guide agricultural practices, help with resource allocation, and contribute to better food production strategies in the face of climate change by looking at historical data and using statistical analysis.

b) Statement of the problem in Technical Term

- This research aims to explore the relationship between climatic variables (temperature, rainfall, relative humidity, and sunshine hours) and rice yield for North-Central Province of Sri-Lanka. The study is considering a two-fold approach by analyzing annual historical data.
 - Utilizing the ARIMA (AutoRegressive Integrated Moving Average) model for forecasting and,
 - Regression Analysis to identify the most influential climatic factor affecting rice yield for North-Central Province of Sri-Lanka.
- Through this analysis of data, the goal is to create a predictive framework for this area that can forecast rice yield based on these climatic variables.

c) Scope of the project

- The research will focus on North-Central Province of Sri-Lanka, by analyzing annual historical data of climatic variables and rice yield from this area. This study considers only weather parameters (temperature, rainfall, relative humidity, and sunshine hours) & rice yield from 1973 to 2023.
- It will also involve identifying the relationship between climatic data & rice yield, developing an ARIMA model to forecast rice yield based on climatic conditions & help to identify the most associated climatic factor for the rice yield.

6. Purpose of Study and Research Objectives

Purpose of the research:

- The purpose of this research is to investigate the effect of climatic variables on rice yield and to develop a forecasting model that can predict it based on these variables for North-Central Province of Sri-Lanka.
- The research aims to provide valuable insights into how climatic factors affect rice cultivation, which can inform agricultural planning, resource allocation, and decision-making processes aimed at enhancing food security and sustainable agricultural practices.

Research Objectives:

- i. To assess the relationship between climatic factors (temperature, rainfall, relative humidity, and sunshine hours) and rice yield.
- ii. To identify the most associated climatic factor for the paddy yield.
- iii. To develop an ARIMA forecasting model for rice yield based on climatic conditions.
- iv. To evaluate the accuracy and reliability of the forecasting models.

7. Significance of the study

- This research exclusively focuses on analyzing weather parameters, specifically annual temperature ($^{\circ}C$), rainfall (mm), relative humidity (%), and sunshine hours.
- The study is specifically focused on the North-Central Province of Sri Lanka.

8. Review of Literature or Similar Project Descriptions

a) Logical flow of literature review

i. Forecasting of Paddy Production in Sri Lanka: A Time Series Analysis using ARIMA Model. (Bogahawatte, 2012)

⇒ This study analyzes paddy production trends in Sri Lanka to aid in planning and import policies. Despite rice self-sufficiency, sector expenditure rises. Objectives include investigating past, present, and future trends, using ARIMA modeling on data from 1952 to 2010. Non-stationarity is addressed by differencing, identifying ARIMA (2, 1, 0) as the best model. The mean Absolute Percentage Error for paddy production is 10.5. Forecasted production for 2011 to 2013 fluctuates, with 2013 surpassing 2010 levels. Continuous model updating with recent data is recommended.

ii. Forecasting Paddy Production in Yala and Maha Seasons in Sri Lanka. (Karunanayaka KJ, 2019)

⇒ This study focuses on forecasting paddy production in Sri Lanka's Yala and Maha seasons using data from 1952 to 2015. ARIMA (3,1,1) and ARIMA (2,1,1) models were found to be the best fits for Yala and Maha seasons, respectively. Additionally, the study investigates the impact of climatic factors such as rainfall, temperature, and pH on paddy production in the Kurunegala district from 1971 to 2017, revealing rainfall as the most influential factor.

iii. Forecasting of rice yield based on weather parameters in Kheda district of Gujarat, India. (Satvinder Kour, 2018)

⇒ This study examines the feasibility of predicting rice yield in Kheda district, Gujarat, by combining weather parameters and technological advancements. ARIMAX models are developed for forecasting rice yield at the tillering stage, incorporating key weather variables. The analysis compares different models based on Relative Mean Absolute Prediction Error (RMAPE), Mean Absolute Deviation (MAD), and Root Mean Square Error (RMSE) values.

- iv. **An Application of ARIMAX Model to Examine the Effect of Climatic Factors on the Annual rice Production in Bangladesh.** (Chowdhury, 2023)
- ⇒ The study investigates the influence of climatic factors on annual rice production in Bangladesh from 1981 to 2020. Using an ARIMAX model, it found that mean temperature negatively affects production, while relative humidity has a positive impact. Precipitation has both positive and negative effects. It suggests developing rice varieties adapted to the country's hot, humid climate to mitigate these impacts.
- v. **Forecast of annual paddy production in MADA region using ARIMA (0,2,2) model.** (Aimi Athirah Ahmad, 2017)
- ⇒ The study focuses on forecasting annual paddy production in Malaysia's MADA region from 2016 to 2020 using five ARIMA models and historical data. It reveals an increasing trend in production over 35 years, predicting a production of 1,257,400 tons by 2020. These forecasts are crucial for shaping national policies on food security.
- vi. **Application of ARIMAX Model for Forecasting Paddy Production in Trincomalee District in Sri Lanka.** (B. Yogarajah, 2013)
- ⇒ The focus is on reviving paddy production in Sri Lanka's Trincomalee district post-war. An ARIMAX model, incorporating rainfall data, is utilized to forecast annual paddy production for both seasons. Model validity is assessed using criteria like Adj R², AIC, SBC, and MAPE values. This aids in developing development plans for the rice sector.
- vii. **Potential Yield of World Rice under Global Warming Based on the ARIMA-TR Model.** (Chengzhi Cai, 2022)
- ⇒ This study uses dynamic ARIMA-TR modeling to project world rice yields by 2030, factoring in historic data since 1961. Projections suggest average yields ranging from 4835 kg/ha to 5195 kg/ha and top yields from 10,127 kg/ha to 10,269 kg/ha by 2030. Insights highlight the need for collaborative efforts between high and low yield countries to enhance global production and ensure food security in the face of climate change.

viii. Improved ARIMAX modal based on ANN and SVM approaches for forecasting rice yield using weather variables. (WASI ALAM, 2018)

⇒ This study proposes ARIMAX hybrid models, ARIMAX-ANN and ARIMAX-SVM, to forecast rice yield using weather variables. Residuals from the ARIMAX model are fitted using ANN and SVM due to their nonlinear patterns. The hybrid models outperform the conventional ARIMAX model, with MAPE values of 0.37 and 1.11 compared to 12.18, respectively. Forecasting yield up to 2020 based on rainfall using ARIMAX (0,1,1) model shows promising results.

ix. Climate change impact on rice yield and production risk. (MAN-KEUN KIM, 2009)

⇒ This study examines the relationship between rice yield and weather variables in Korea. Findings indicate a positive correlation between average rice yield and temperature, but a negative correlation with precipitation. With anticipated increases in temperature and precipitation, market risk is expected to grow. Climate change may escalate rice yield variability by 10%~20%. To mitigate these effects, government farm policy and research should focus on reducing yield variability and managing market risk.

x. Statistical modelling for forecasting of wheat yield based on weather variables. (RANJIT KUMAR PAUL, 2013)

⇒ Study employs the ARIMAX time-series model to develop five models corresponding to key stages of wheat growth, incorporating important weather variables. Evaluation based on Relative Mean Absolute Prediction Error (RMAPE) shows the effectiveness of the ARIMAX methodology. Forecasts become more accurate as the wheat crop matures, with predictions aligning closely with actual values closer to harvest.

xi. Rainfall forecasting with climate change detection and its pattern relationship to rice production. (Endah R.M. Putri)

⇒ The study utilizes SARIMA modeling to predict rainfall intensity, crucial for agricultural planning amid climate change. The Copula method explores the relationship between rainfall intensity and its impact on rice production. Findings suggest SARIMA ([1, 2, 6], 1, 0) (0, 1, 1) as the optimal rainfall forecasting model, with indications of climate change in rainfall intensity in Tabanan, Bali.

- xii. Forecasting of Paddy Grain and Rice's Price: An ARIMA (Autoregressive Integrated Moving Average) Model Application.** (Fitri Ramadhani, 2020)
- ⇒ This study employs ARIMA models to forecast paddy grain and rice prices at various market levels in Indonesia. The chosen ARIMA models are based on the lowest AIC and SC accuracy values. For paddy grain and rice prices, different ARIMA models are selected for farmer, milling, wholesale, and international levels. The forecasted prices closely match the actual prices, indicating the effectiveness of the ARIMA models in price prediction.
- xiii. Application of ARIMA Model for Forecasting Agricultural Productivity in India.** (PADHAN, 2012)
- ⇒ This study explores forecasting agricultural crop productivity in India using ARIMA models. Considers factors such as rainfall, fertilizer use, and subsidies. ARIMA models are applied to annual data from 1950 to 2010 for 34 different crops. Model validity is assessed using criteria like Adj R², AIC, and MAPE. Tea and cardamom exhibit the most accurate forecasts, with low MAPE and AIC values, respectively.
- xiv. Effects of Climate Change on the Yield and Cropping Area of Major Food Crops: A Case of Bangladesh.** (Md. Ruhul Amin, 2015)
- ⇒ Examines the impact of climate change on four major food crops in Bangladesh. Data from 1972 to 2010, it finds significant effects of climate variables on crop yield and cropping area. Maximum temperature adversely affects all crop yields except Aus rice, while rainfall significantly affects Aman rice. Authorities should focus on developing drought and flood-tolerant varieties to ensure food security amidst changing climate conditions.
- xv. ARIMA forecast of area, production, and productivity of rice in amalgamated Raipur using R.** (SINGH, 2018)
- ⇒ This study focuses on forecasting rice production in Raipur district, using time series data from 1980 to 2015. ARIMA and double exponential methods were employed for forecasting, with models developed using the R software. Good models were identified for area, production, and productivity, with forecasts made for the period from 2015-16 to 2019-20 based on these models.

b) Justification on relevance of the literature for the project

Journal Number	Geographical Area	Objectives	Methodology
i.	Sri Lanka.	- To analyze trends in paddy production. - To investigate past, present, and future trends.	Utilized ARIMA modeling on paddy production data from 1952 to 2010, addressing non-stationarity through differencing.
ii.	Kurunegala District, Sri-Lanka	- To forecast paddy production in both Yala and Maha seasons. - To analyze the impact of climatic factors (rainfall, temperature, pH)	Utilized ARIMA modeling to forecast paddy production in Yala and Maha seasons and investigated the relationship between climatic factors and paddy production using data from 1952 to 2015 for production and 1971 to 2017 for climatic factors in the Kurunegala district.
iii.	Kheda district, Gujarat, India.	To forecast rice yield by considering weather parameters.	Developed ARIMAX models to predict rice yield based on weather variables. Models were compared based on RMAPE, MAD, and RMSE values.
iv.	Bangladesh.	To examine the impact of climatic factors on annual rice production	Applied an ARIMAX model to analyze the relationship between climatic factors and rice production using data from 1981 to 2020.
v.	MADA region, Malaysia.	To forecast annual paddy production.	Utilized five ARIMA models and historical data to forecast paddy production from 2016 to 2020.

vi.	Trincomalee District, Sri Lanka.	- To revive paddy production post-war. - To forecast annual production for both (yala & maha) seasons.	Utilized an ARIMAX model incorporating rainfall data to forecast paddy production, assessed model validity using criteria like Adj R ² , AIC, SBC, and MAPE values.
vii.	Global (Worldwide).	To predict world rice yields by 2030 under global warming conditions.	Utilized dynamic ARIMA-TR modeling based on historic data since 1961 to project average and top yields of world rice by 2030.
viii.	Aligarh district of Uttar Pradesh, India.	To improve the forecasting accuracy of rice yield using weather variables	Developed ARIMAX hybrid models by fitting residuals from the ARIMAX model using ANN and SVM. Compared the performance of hybrid models with the conventional ARIMAX model using MAPE values.
ix.	Korea.	- To identify the relationship between rice yield and weather variables - To assess the impact of climate change on rice yield variability and production risk.	Examined historical data to determine the correlation between rice yield and weather variables in Korea. Anticipated future changes in temperature and precipitation were considered to forecast potential increases in rice yield variability.
x.	Kanpur district of Uttar Pradesh, India.	To forecast wheat yield based on weather variables.	Developed five ARIMAX models corresponding to key stages of wheat growth, integrating important weather variables. Evaluated model effectiveness using RMAPE. Forecasts were made for different stages of wheat growth, with accuracy improving as the crop matures.

xi.	Tabanan, Bali province, Indonesia.	- To forecast rainfall intensity - To analyze its relationship with rice production, by considering climate change.	Utilized SARIMA model to examine the relationship between rainfall intensity and rice production. Identified SARIMA ([1, 2, 6], 1, 0) (0, 1, 1) as the optimal forecasting model and detected indications of climate change in rainfall intensity.
xii.	Indonesia.	- To forecast paddy Grain Price on the Farmer Level. - To forecast Paddy Grain Price on the Milling Level. - To forecast Rice Price on the Wholesale Level.	Utilized ARIMA models based on the lowest AIC and SC accuracy values for forecasting paddy grain and rice prices. The effectiveness of the models were assessed by comparing the forecasted prices with actual prices.
xiii.	India.	To forecast agricultural crop productivity in India based on rainfall, fertilizer use, and subsidies.	Applied ARIMA models to annual data for 34 different crops. Model validity assessed using Adj R ² , AIC, and MAPE. Tea and cardamom selected as the focus crops due to their accurate forecasts with low MAPE and AIC values, respectively.
xiv.	Bangladesh.	To examine the impact of climate change on the yield.	Utilized data from 1972 to 2010 to assess the effects of climate variables on crop yield. ADF test for checking the stationarity. Performing Descriptive & Regression Analysis.
xv.	Raipur district, India.	To forecast rice production area, and productivity within this region.	Utilized ARIMA and double exponential methods for forecasting, using R software.

Table 1; Justification on relevance of the literatures.

c) Chapter conclusion with remarks

In here, I 'm going to do the research related on the “Effect of climatic variables on rice yield and its forecast in the North-Central Province of Sri Lanka”. Through a comprehensive review of relevant literatures, I have gained insights into the complex relationship between climatic factors and rice yield, drawing parallels between studies conducted in similar agricultural researches.

My examination of existing research, particularly the studies conducted in several countries/regions/provinces have provided valuable methodological frameworks and findings that inform me, how understanding of the impact of temperature, humidity, rainfall, and sunshine hours on rice cultivation. By leveraging statistical analyses and modeling techniques such as the ARIMA model, I hope to forecast rice yield based on climatic variables, thereby contributing to predictive agriculture practices in the region.

Remarks:

- The literature review serves as a solid foundation for my research. They are guiding my methodology and approach towards investigating the impact of climatic variables on rice yield in the North-Central Province of Sri Lanka.
- Insights gleaned from studies conducted in analogous agricultural settings underscore the significance of considering diverse climatic factors and their temporal variations in predicting rice yield.
- The utilization of statistical analyses and modeling techniques demonstrated in the literature reviews, particularly the ARIMA model. They provide a robust framework for forecasting rice yield based on climatic variables, enhancing the predictive capacity of my research.
- Building upon the findings and methodologies outlined in the literature, my aim to offer fresh perspectives on the complex relationship between climatic variables and rice production in Sri Lanka's North-Central Province. By doing so, it seeks to support the development of sustainable agricultural practices and ensuring food security in the region.

9. Methodology (design, methods, procedures)

i. Research Design:

- Utilizing a longitudinal study design to observe changes in rice yield over time in relation to climatic variations.

ii. Objectives:

- To assess the relationship between annual climatic factors and rice yield.
- To identify the most associated climatic factor for the paddy yield.
- To develop an ARIMA forecasting model for rice yield based on climatic conditions.
- To evaluate the accuracy and reliability of the forecasting model.

iii. Data Collection Methods:

- Considering data collection period from 1973 to 2023.
- Collecting annual historical data on rice yield in the North-Central Province from relevant government agencies, agricultural research institutions, or some valuable online databases such as Department of Census and Statistics (DCS, n.d.), Ministry of Agriculture & Plantation Industries. (Ministry of Agriculture and Plantation Industries, n.d.)
- Obtaining climatic data including temperature, rainfall, humidity, and sunshine hours from meteorological stations, relevant authorities, or some valuable online databases for the same period. (meteoblue, n.d.) (Weather and Climate, n.d.) (Climate Data, n.d.)

iv. Variables and Measures:

- Identifying annual key climatic variables such as average temperature ($^{\circ}\text{C}$), rainfall(mm), humidity (%), and sunshine hours as independent variables.
- Define annual rice yield as the dependent variable, measured in metric tons.

v. Data Cleaning & Preprocessing:

- Identifying missing values and outliers. Then must clean data to do the analysis part.

vi. Statistical Analysis:

- Conducting Descriptive Analysis to analyze the distribution and variability of rice yield and climatic variables.
- Correlation and Regression Procedures used to identify mostly affected climatic factor.
- Performing ARIMA modeling, to build up forecast rice yield.

vii. Forecasting Methodology:

- Developing forecasting models based on historical trends and patterns observed in rice yield and climatic variables.
- Applying time series analysis techniques, such as ARIMA models, to forecast future rice yield based on anticipated climatic conditions.

viii. Model Validation & Testing:

- This will involve comparing the predicted rice yield values generated by the models with actual observed values from the historical dataset.
- Validate the forecasting models using cross-validation techniques and assess their accuracy and reliability.
- Then we can accurately predict rice yield under different climatic scenarios.

ix. Results & Discussion:

- It will discuss the significance of these relationships and their implications for agricultural practices in the North-Central Province of Sri Lanka.
- It will identify most associated climatic factor for the rice yield.
- Also, can develop a best accurate & reliable ARIMA forecasting model for rice yield based on climatic conditions for this area.

10. Work plan and Implications (Gantt chart)

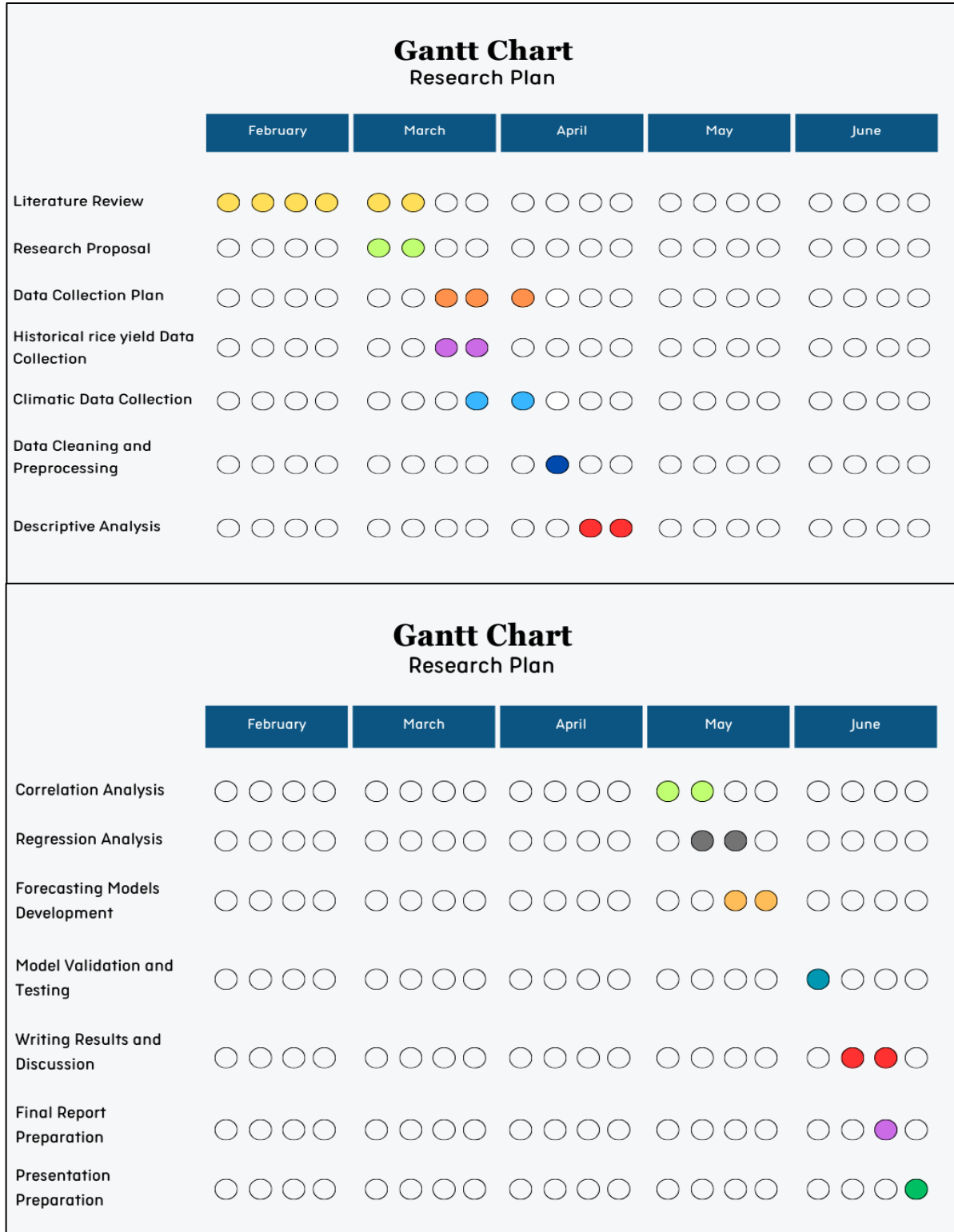


Figure 3; Gantt Chart for Research Plan.

11. Limitations & Delimitations

❖ Limitations:

Factor	Description
Geographical Scope	✓ The research is limited to the North Central Province of Sri-Lanka. ✓ Findings and conclusions will be applicable only within this geographical area and may not be generalizable to other regions or provinces.
Climatic Variables	✓ Consider only annual data parameters. ✓ The study will focus specifically on the following climatic variables such as: temperature, relative humidity, sunshine hours, and rainfall. ✓ Other climatic factors may impact rice yield but will not be included in the analysis
Data Collection Time period	✓ Considering only from 1973 to 2023.
Data Sources	✓ Data for the research will be obtained from reliable sources such as government agencies, agricultural research institutions, or reputable online databases.

Table 2; Limitations of the Research

❖ Delimitations:

Factor	Description
Geographical Scope	✓ The research is valid for any places within the North Central Province of Sri-Lanka. ✓ Findings and conclusions will be valid for any places within this geographical area.

Table 3; Delimitations of the Research

12. Conclusion

⇒ Effect of Climatic Variables on Rice Yield:

- The research findings may reveal significant relationships between climatic annual data (temperature, relative humidity, sunshine hours, and rainfall) and rice yield in the North Central Province.

⇒ Seasonal Variability in Rice Yield:

- The analysis may highlight seasonal variations in rice yield in response to fluctuations in climatic conditions throughout the growing period.
- Certain climatic variables may have stronger correlations with rice yield during specific seasons.

⇒ Can identify the most associated climatic factor for the rice yield.

⇒ Forecasting Accuracy:

- The forecasting models developed in the study using ARIMA models, may demonstrate varying levels of accuracy in predicting future rice yield based on anticipated climatic conditions.
- The research may provide insights into the reliability and limitations of the forecasting models in capturing the complex interactions between climatic variables and rice production dynamics.

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